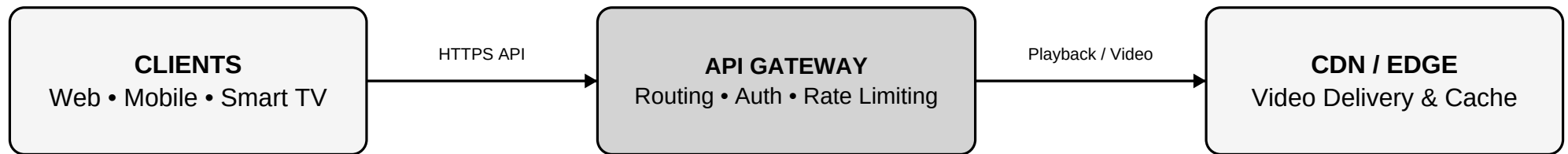


# Netflix High-Level Design (HLD)

Professional Draw.io-style architecture | Clean flow and separated data/video planes

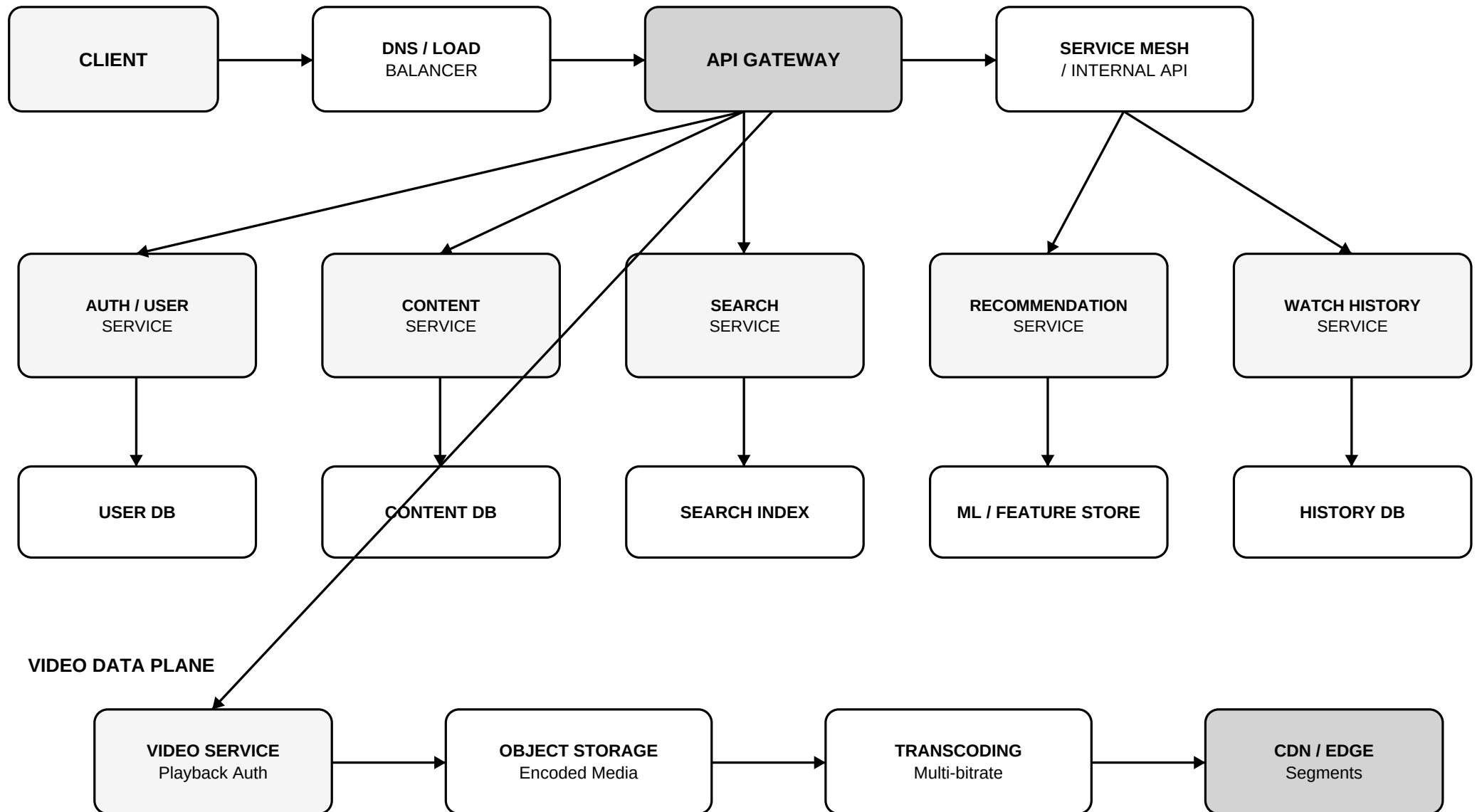


## Core principle

Keep high-volume video traffic on the CDN path and keep business APIs on the service path.

# Netflix HLD — Main Architecture

All service boxes are aligned to avoid crossing connections



API/control plane → metadata & authorization | CDN data plane → video bytes

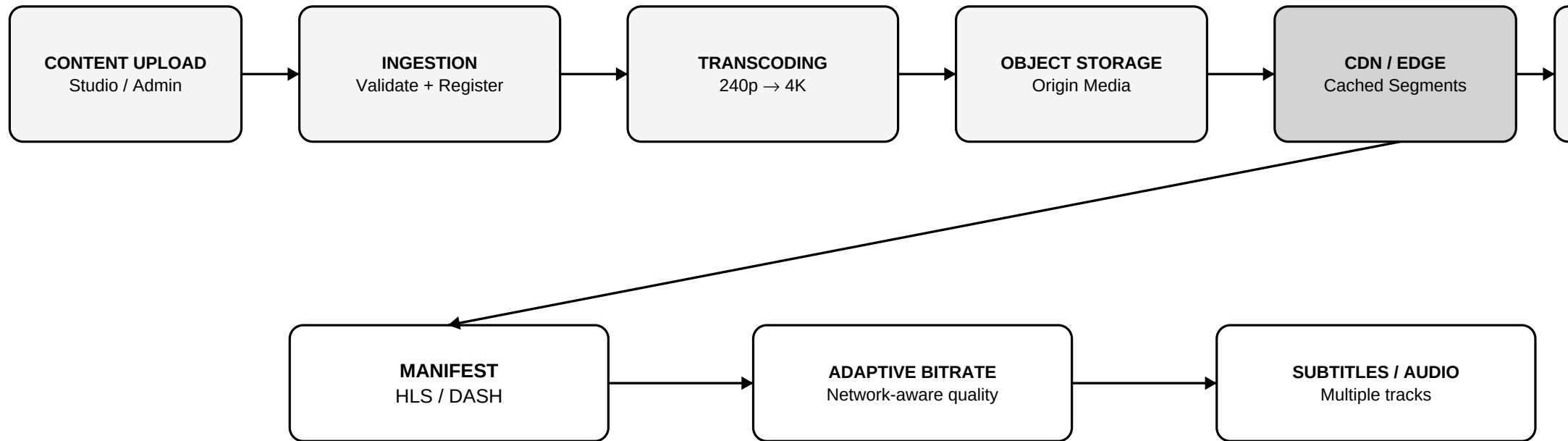
# Netflix HLD — Service Responsibilities

Logical microservices and their primary responsibilities



# Netflix Video Pipeline

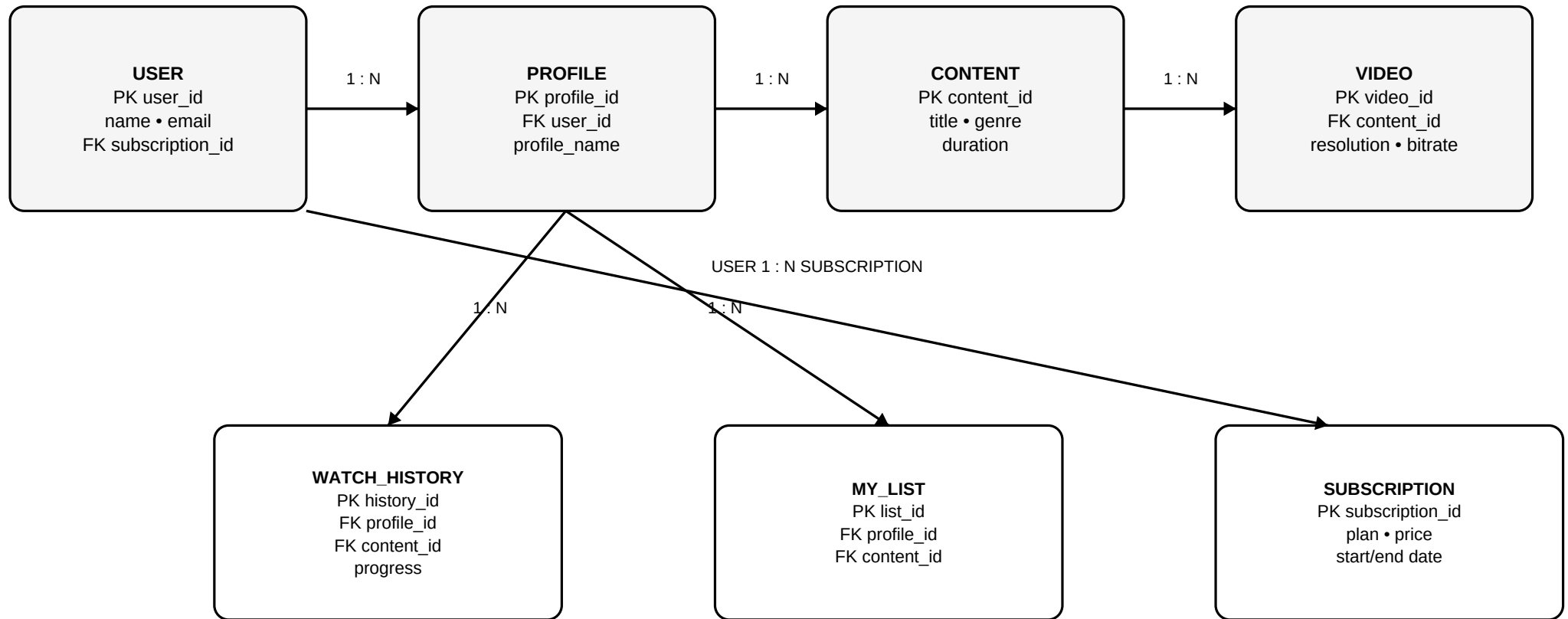
Upload → encode → store → distribute → adaptive playback



Playback: client requests manifest → CDN serves small segments → player changes bitrate based on bandwidth and buffer.

# Netflix ER Diagram

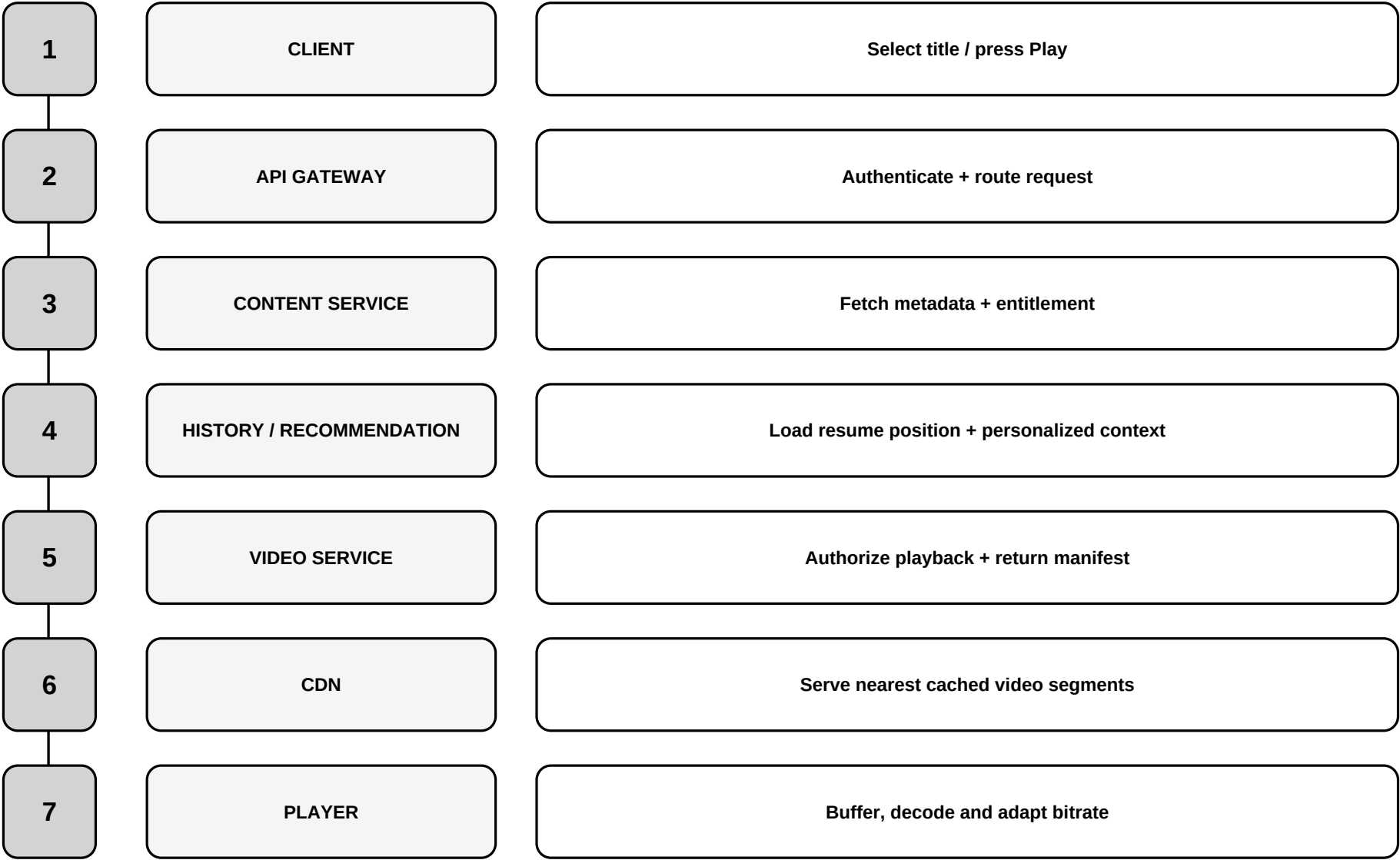
Core relational/logical data model



A profile can watch many titles; a title can appear in many histories/lists. CONTENT has multiple encoded VIDEO variants.

# Netflix — End-to-End Request Flow

Example: user opens a title and starts playback



**RESILIENCE**  
Timeouts • Retries  
Circuit Breakers  
Rate Limiting

**OBSERVABILITY**  
Logs • Metrics  
Tracing • Alerts

**Design priorities** Scalability • High availability • Low latency • Fault tolerance • Secure playback • Reliable streaming